Achilles Tendon Load is Progressively Increased with Reductions in Walking Speed

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ABSTRACT

BRAUNER, T., P. POURCELOT, N. CREVIER-DENOIX, T. HORSTMANN, and S. C. WEARING. Achilles Tendon Load is Progressively Increased with Reductions in Walking Speed. *Med. Sci. Sports Exerc.*, Vol. 49, No. 10, pp. 2001–2008, 2017. **Introduction**: Achilles tendon rehabilitation protocols commonly recommend a gradual increase in walking speed to progressively intensify tendon loading. This study used transmission-mode ultrasound to evaluate the influence of walking speed on loading of the human Achilles tendon *in vivo*. **Methods**: Axial transmission speed of ultrasound was measured in the right Achilles tendon of 33 adults (mean ± SD: age, 29 ± 3 yr; height, 1.725 ± 0.069 m; weight, 71.4 ± 19.9 kg) during unshod, steady-state treadmill walking at three speeds (slow, 0.85 ± 0.12 ms; preferred, 1.10 ± 0.13 $m \cdot s^{-1}$; fast, 1.35 ± 0.20 $m \cdot s^{-1}$). Ankle kinematics, spatiotemporal gait parameters and vertical ground reaction force were simultaneously recorded. Statistical comparisons were made using repeated-measures ANOVA models. **Results**: Increasing walking speed was associated with higher cadence, longer step length, shorter stance duration, greater ankle plantarflexion, higher vertical ground reaction force peaks, and a greater loading rate ($P < 0.05$). Maximum ($F_{1,38} = 7.38$, $P < 0.05$) and minimum ($F_{1,46} = 8.95$, $P < 0.05$) ultrasound transmission velocities in the Achilles tendon were significantly lower (16–23 $m \cdot s^{-1}$) during the stance but not swing phase of gait, with each increase in walking speed. **Conclusions**: Despite higher vertical ground reaction forces and greater ankle plantarflexion, increasing walking speed resulted in a reduction in the axial transmission velocity of ultrasound in the Achilles tendon; indicating a speed-dependent reduction in tensile load within the triceps surae muscle–tendon unit during walking. These findings question the rationale for current progressive loading protocols involving the Achilles tendon, in which reduced walking speeds are advocated early in the course of treatment to lower Achilles tendon loads. **Key Words:** ULTRASOUND, SPEED OF SOUND, LOCOMOTION, ACHILLES TENDINOPATHY, SOFT TISSUE, BIOMECHANICS

Progressive loading is widely advocated in the rehabilitation of Achilles tendon injuries. In particular, slow walking is specifically recommended early in the management of Achilles tendon disorders to lower tendon force, with progressively faster gait speeds recommended later in the course of treatment to increase tendon loading as pain allows (35). The effect of walking speed on loading of the Achilles tendon, however, is controversial, with faster walking speeds suggested to either increase, decrease, or have no effect on Achilles tendon loading (4,11,13,19,25,26).

Although some modelling studies of human walking have suggested that increasing gait speed requires greater work from the ankle plantarflexors (26), similar musculoskeletal simulations have proposed that faster walking speeds increase the shortening velocity of triceps surae muscle fibers, which impairs ankle plantarflexion force production (25). In Neptune and Sasaki's simulations, muscle activation and force increased with walking speed in all lower limb muscle groups evaluated, except the triceps surae muscle, in which force decreased at walking speeds faster than 80% of the walk–run transition speed (i.e., ≈1.57 $m \cdot s^{-1}$) (25). In support of this concept, Farris and Sawicki (11) used a combination of inverse dynamics and ultrasound measurements to show experimentally that peak force in the medial gastrocnemius muscle was lower when walking at 2.0 $m \cdot s^{-1}$ than at 1.25 $m \cdot s^{-1}$. Interestingly, on closer examination, even slow walking (0.75 $m \cdot s^{-1}$) resulted in a lower peak force in the medial gastrocnemius muscle compared to walking at 1.25 $m \cdot s^{-1}$, despite inducing a slower muscle fiber–shortening velocity. An inherent limitation of inverse-dynamics approach, however, is the need to estimate the muscle–tendon moment arm and the inability to devolve the individual contribution of muscles to joint mechanics. In a more recent musculoskeletal simulation in which the relative activation of the ankle plantarflexors was based on electromyographic activity, Arnold et al. (4) demonstrated that despite the triceps surae muscle suffering from less favorable contractile conditions

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Submitted for publication December 2016.
Accepted for publication May 2017.

0195-9131/17/4910-2001/0
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DOI: 10.1249/MSS.0000000000001322

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at faster walking speeds, plantarflexor force was unchanged; thereby highlighting the need for more direct measures of tendon loading.

Animal models in which tendon forces were directly measured via surgically implanted transducers, have shown that despite a speed-dependent reduction in contractile ability of the gastrocnemius and soleus muscles, peak force within their respective tendons was either increased or unchanged with increasing walking speed (16,17,30). Although limited in their sample size, analogous studies involving direct measurement of human Achilles tendon during walking have shown that peak force in the Achilles tendon also remains relatively unchanged with increasing walking speed. Of the two studies undertaken to date, it is noteworthy that one showed a trend for a small decrease in peak force in the Achilles tendon (19), whereas the other showed a small increase in peak force at walking speeds ranging between 1.1 and 1.8 $m \cdot s^{-1}$ (13). Although such studies have advanced our understanding of tendon function with increasing locomotor speed, direct measurement of the Achilles tendon is too invasive for routine use in humans and, as such, to date has been limited to relatively small study populations.

Transmission-mode ultrasound, in contrast, has the potential to provide direct, though noninvasive, estimates of human tendon loading *in vivo,* and is suitable for application in large participant cohorts. The operational principle of the transmission technique is described in detail elsewhere (7). In brief, the axial transmission velocity of ultrasonic waves in tendon (S) is dependent on the instantaneous elastic modulus (E) and density (ρ) of tendon and is governed by the classic Newtonian–Laplace equation with some adjustment for Poisson effects (ν) in elastic media (21,24,28,36):

$$S = \sqrt{\frac{E}{\rho}\frac{(1-\nu)}{(1+\nu)(1-2\nu)}}$$

Studies in animal tendon have confirmed that the transmission speed of axial ultrasonic waves is also proportional to the applied tensile load in tendon (21,24,28). Recently, ultrasound transmission techniques have been shown to be sensitive to changes in tendon loading with footwear (38), orthotic intervention (40), tendon injury (37), and between walking and slow running gaits (41). The aim of this study, therefore, was to use transmission-mode ultrasound to evaluate the influence of walking speed on loading of the human Achilles tendon *in vivo*. In particular, it was hypothesized that ultrasound velocity in the Achilles tendon would increase with advancing walking speed; reflecting a speed-dependent increase in tensile load within the muscle–tendon unit.

METHODS

Participants. Thirty-three healthy adults (16 female, 17 male) were recruited from university faculty to participate in the study. The mean (±SD) age, height, and body mass of participants was 29 ± 3 yr, 1.725 ± 0.069 m, and 71.4 ± 19.9 kg, respectively. Participants were nonsmokers, nonmedicated, and recreationally active based on self-report. No participant reported a medical history of diabetes, inflammatory arthritis, familial hypercholesterolemia, or Achilles tendon injury or pathology. Participant numbers were based on previously published data for human Achilles tendon (38) and were sufficient to detect a 3% difference in the peak propagation velocity of ultrasound in the tendon ($\alpha = 0.05$, $\beta = 0.20$). All participants gave written informed consent before participation in the research, which received approval from the University Human Research Ethics Committee.

Equipment. A flexible strain-gauge electrogoniometer (Penny and Giles, Biometrics, Gwent, UK) was used to measure sagittal plane ankle movement during walking. The endblocks of the device were attached to the skin overlying the medial calcaneus and the distal aspect of the medial tibia of the right ankle using double-sided adhesive tape. Once secured, the goniometer was zeroed during erect bipedal stance. The electrogoniometer has a resolution of 1°, and has been shown to be accurate to within 1.5% over a measurement range of 100° (32).

Vertical ground reaction force and spatiotemporal gait parameters were determined during gait using an instrumented treadmill system (Zebris Medical GmbH, Isny, Germany). The system is composed of a capacitance-based pressure platform housed within a motor-driven treadmill. The pressure platform had a sensing area of 108.4 × 47.4 cm and incorporated 7168 sensors, each approximately 0.85 × 0.85 cm. The treadmill had a contact surface of 150 × 50 cm and its belt speed could be adjusted between 0.2 and 22 $km \cdot h^{-1}$, at intervals of 0.1 $km \cdot h^{-1}$. The coefficients of variation for spatiotemporal gait parameters measured during walking are typically <10% for repeated measures (31).

Axial transmission velocity of ultrasound was measured in the right Achilles tendon using a custom-built ultrasonic device similar to that described previously (8,28,37,38). The probe consisted of a 1-MHz broadband pulse emitter and four regularly spaced receivers and was positioned over the midline of the posterior aspect of the Achilles tendon, with the emitter located approximately 1 cm above the calcaneus and maintained in close contact with the skin by means of coupling medium and elasticized straps (Fig. 1). Received ultrasonic signals were digitized at 20 MHz and the time of flight of the first arriving transient between receivers was estimated using the first zero crossing criterion. Average transmission velocity of ultrasound was subsequently calculated given the known distance between pairs of receivers and the measured time of flight. Measurement precision for axial transmission velocity of ultrasound is better than 3 $m \cdot s^{-1}$, and the error in predicting applied tensile force in tendon from direct measures of ultrasound velocity is typically <2% (8).

Protocol. Participants reported to the gait laboratory wearing lightweight, comfortable clothing, and having abstained from vigorous physical activity on the day of testing. Before the fixation of the ultrasound probe and electrogoniometer to the right

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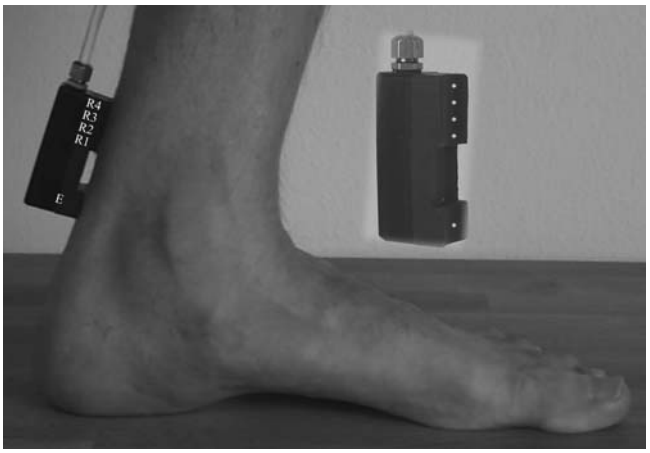

FIGURE 1—Axial transmission velocity of ultrasound in the right Achilles tendon was measured with a custom built ultrasonic probe, which incorporated a 1-MHz emitter (E) and four receivers (R1-4). The probe was positioned over the posterior aspect of the Achilles tendon, with the emitter located approximately 1 cm superior to the proximal border of the calcaneum. An adhesive coupling medium was applied to the contact surface of the probe (*) (inset). Elasticated straps have been removed for illustrative purposes.

leg, the skin overlying the posterior Achilles tendon and medial aspect of the right shank was prepared and cleaned using standard alcohol abrading methods. Participants were then afforded a treadmill acclimatization session which included a 6-min period of steady-state walking at a self-selected "comfortable" speed. The "comfortable" walking speed of each participant was determined at the outset of the acclimatization session using a previously outlined method (31). In brief, self-selected "comfortable" walking speed was determined by gradually increasing the belt speed until each participant indicated he/she had reached his/her natural pace. The treadmill speed was then increased by 1.0 $km \cdot h^{-1}$ and, blinded to the participant, reduced in 0.1 $km \cdot h^{-1}$ steps until the participant perceived that his/her preferred "comfortable" walking pace had been reestablished (31). This speed was subsequently recorded as the participant's preferred walking speed.

After acclimatization, participants randomly completed three barefoot gait conditions; walking at slow (≈−25% preferred), preferred and fast (≈+25% preferred) gait speed. After 6 min of steady-state gait at each speed, ultrasound velocity in the right Achilles tendon, sagittal plane ankle movement, vertical ground reaction force, and basic spatiotemporal gait data were synchronously sampled for 60 s at a rate 120 Hz. Approximately 44 ± 4, 50 ± 4, and 56 ± 4 gait cycles were analyzed for each participant at slow, preferred, and fast walking speeds, respectively.

Data reduction and statistical analysis. Force, time, and velocity data were exported from the treadmill system in ASCII format and custom computer code (Matlab R2012a, MathWorks, Natick, MA) was subsequently used to determine spatiotemporal parameters, including cadence, stride duration, step length, stance, and swing phase durations. The magnitude and timing of peak vertical ground reaction forces during walking and running were calculated (31,39), together with the peak loading rate (PLR), which was defined as the peak instantaneous force differential during the stance phase of gait. Local maximum dorsiflexion (DF1, DF2) and plantarflexion (PF1, PF2) of the ankle joint during each gait cycle were identified, and dorsiflexion was represented by positive values. Similarly, local maxima (P1, P2) and minima (M1, M2) in ultrasound transmission velocity were analyzed over the gait cycle. Values were derived for each gait cycle recorded within the 60-s data capture period and the mean values for all gait cycles were calculated and used in subsequent statistical analysis.

The Statistical Package for the Social Sciences (Version 23, IBM Corp. Armonk, NY) was used for all statistical procedures. Shapiro Wilkes tests were used to evaluate data for underlying assumptions of normality. Outcome variables were determined to be normally distributed and therefore means and SD have been used as summary statistics. Potential differences in the vertical ground reaction force, spatiotemporal gait parameters, ankle joint kinematics, and the velocity of ultrasound propagation in the Achilles tendon with varying gait speeds were assessed using repeated-measures ANOVA within a generalized linear modelling framework. In each case, speed (slow, preferred, fast) was treated as a within–subject factor. Underlying assumptions regarding the uniformity of the variance–covariance matrix were assessed using Mauchly test of sphericity. When the assumption of uniformity was violated, an adjustment to the degrees of freedom of the F ratio was made using Greenhouse–Geisser Epsilon, thereby making the F-test more conservative. Where significant differences were identified, planned contrasts were undertaken. Effect sizes were estimated via the partial eta statistic. An alpha level of 0.05 was used for all tests of significance.

RESULTS

Basic temporospatial gait parameters at slow, preferred, and fast walking speeds are shown in Table 1. All participants adopted a rearfoot footstrike pattern during barefoot walking at each speed as determined from visual inspection of pressure–time data from the treadmill system. Faster walking speeds were characterized by a longer stride duration ($F_{2,64}$ = 1165.5, P < 0.05), greater step length ($F_{2,64}$ = 270.8, P < 0.05), shorter stance phase duration and shorter initial ($F_{1,46}$ = 243.7, P < 0.05) and final ($F_{2,49}$ = 284.8, P < 0.05) periods of double support than slow walking.

Accordingly, peak vertical ground reaction forces were higher (F1, $F_{1,37}$ = 27.1, P < 0.05; F3, $F_{1,44}$ = 16.2, P < 0.05) and occurred earlier (TF1, $F_{2,52}$ = 103.3, P < 0.05; TF3, $F_{1,64}$ = 58.3, P < 0.05) in stance with increasing walking speed (Fig. 2). The local minimum (F2), in contrast, progressively decreased with increasing speed ($F_{1,40}$ = 250.0, P < 0.05). Peak loading rate also increased with walking speed ($F_{1,40}$ = 355.9, P < 0.05), with the loading rate at the fast walking speed approximately twice that of the slow walking condition (Table 2).

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TABLE 1. Mean (SD) temporospatial gait characteristics of participants at slow, preferred, and fast walking speeds.

	Slow	Preferred	Fast	η_p^2	P
n	33	33	33		
Walking speed (m·s^{-1})	0.85 (0.12)	1.10 (0.13)	1.35* (0.20)	0.96	<0.001
Cadence (strides per second)	0.79 (0.05)	0.91 (0.06)	1.00* (0.07)	0.97	<0.001
Stride duration (s)	1.27 (0.08)	1.10 (0.07)	1.01* (0.07)	0.97	<0.001
Stance phase duration (%GC)	65.1 (1.5)	63.5 (1.6)	62.0* (1.7)	0.87	<0.001
Swing phase duration (%GC)	34.9 (1.5)	36.5 (1.6)	38.0* (1.7)	0.87	<0.001
Initial period of double support (%GC)	15.5 (1.6)	13.9 (1.4)	12.3* (1.5)	0.88	<0.001
Final period of double support (%GC)	15.3 (1.4)	13.7 (1.4)	12.2* (1.5)	0.90	<0.001
Step length (m)	0.70 (0.07)	0.74 (0.06)	0.83* (0.08)	0.89	<0.001

*Statistically significant difference between all walking speeds ($P < 0.05$).
%GC, percentage of gait cycle; η_p^2, partial eta squared.

Table 3 summarizes the effect of walking speed on sagittal motion of the ankle. A small but statistically significant decrease in maximum dorsiflexion (DF1, $F_{1,41} = 10.4$, $P < 0.05$) and an increase in maximum plantarflexion was observed during initial contact (PF1; $F_{1,47} = 7.5$, $P < 0.05$) and swing (PF2; $F_{1,42} = 8.9$, $P < 0.05$) with each increase in walking speed (Fig. 2).

The pattern of ultrasound transmission velocity in the Achilles tendon was highly reproducible for all walking speeds (Fig. 2), with a mean within-subject coefficient of variation for ultrasonic maxima and minima ranging between 0.2% ± 0.1% and 1.0% ± 0.6%. For all speeds, ultrasound transmission velocity in the Achilles tendon decreased immediately after heel strike (M1, ≈10% of the gait cycle), to peak in late stance (P1, ≈50% of the gait cycle) before reaching a second minimum value during the early swing (M2, ≈70% of the gait cycle). Walking speed had a significant effect on the first ultrasound transmission velocity minimum (M1, $F_{1,46} = 8.9$, $P < 0.05$); with slow walking producing a higher transmission velocity in the Achilles tendon than preferred walking (+16 m·s^{-1}, $P < 0.05$), which in turn, was higher than that of the faster walking speed (+23 m·s^{-1}, $P < 0.05$). Similarly, walking speed also had a significant effect on the first peak in ultrasound transmission velocity of the Achilles tendon (P1, $F_{1,38} = 7.4$, $P < 0.05$); with slower speed producing a higher transmission velocity in the Achilles tendon than preferred walking (+23 m·s^{-1}, $P < 0.05$), which was successively higher than that of the fast walking speed (+20 m·s^{-1}, $P < 0.05$). There was no significant effect of gait speed on maximum (P2) and minimum (M2) transmission velocity of ultrasound in the Achilles tendon during the swing phase of gait (Table 4).

DISCUSSION

This study used transmission-mode ultrasound to evaluate the influence of walking speed on loading of the human Achilles tendon *in vivo*. Contrary to our hypothesis, we observed a progressive reduction in the axial transmission velocity of ultrasound in the Achilles tendon with increasing walking speed, indicating a speed-dependent reduction in tensile load and/or loading rate within the triceps surae muscle–tendon unit.

Peak transmission velocity of ultrasound in the Achilles tendon in our study was comparable to that reported previously for human tendon (≈1900–2050 m·s^{-1}) (28,29), and was of a similar pattern to the biphasic force profile reported for the Achilles tendon during overground walking when measured directly with implanted force transducers (13,19). However, in contrast to the latter studies in which peak Achilles tendon force was found to be relatively invariant to changes in walking speed between 1.1 and 1.8 m·s^{-1} (13,19), we observed a progressive decrease in axial transmission velocity of ultrasound in the Achilles tendon over walking speeds of 0.85 to 1.35 m·s^{-1}. The observation of a speed-dependent reduction in tensile load within the triceps surae muscle–tendon unit during walking in the current study raises two scenarios for further discussion.

First, internal loading of a muscle–tendon unit has been suggested to be dependent on both the amplitude of joint movement and contraction intensity of the corresponding muscle. Consistent with previous research, we observed a concomitant increase in peak vertical ground reaction forces and a small but progressive decrease in ankle dorsiflexion and increase in ankle plantarflexion with increasing walking speed from 0.85 to 1.35 m·s^{-1} (4,27). It is possible, therefore, that the speed-induced reduction in ultrasound transmission velocity in the Achilles tendon reflects a lengthening of the tendon moment arm which reportedly accompanies ankle plantarflexion (23). The effect of ankle movement on the length of the Achilles tendon moment arm, however, is controversial; with ankle plantarflexion noted to increase, decrease or have no effect on Achilles moment arm length (15,23,34). Similarly, assuming that reduced ankle dorsiflexion at faster walking speed results in a lower ground reaction force moment arm and lower external ankle dorsiflexion torque, it is also possible that the reduction in transmission velocity of ultrasound in the tendon reflects reduced muscle activity in the ankle plantarflexors associated with a lower external ankle

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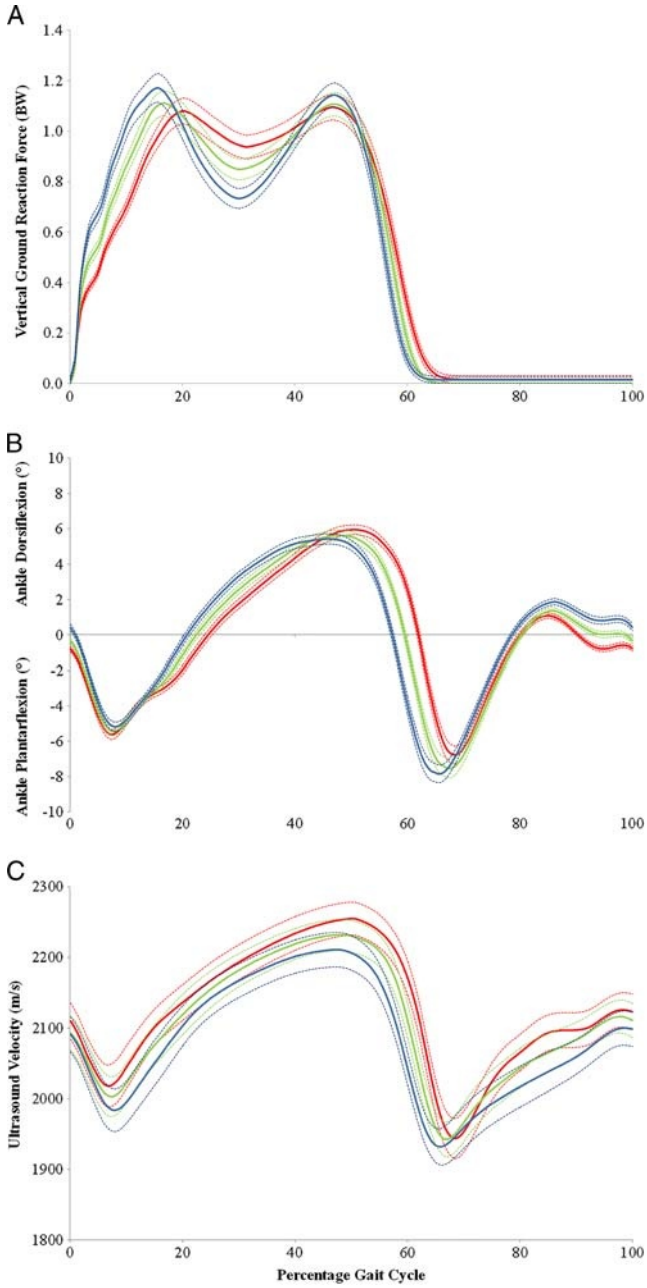

FIGURE 2—Ensemble histories for vertical ground reaction force (A), sagittal plane ankle movement (B), and ultrasonic velocity in the right Achilles tendon (C) during barefoot walking at preferred (*green line*), slow (*red line*, −23% preferred), and fast (*blue line*, +23% preferred) walking speeds. Ensembles represent the average of 44 ± 4, 50 ± 4, and 56 ± 4 gait cycles from each participant (*n* = 33) walking at slow, preferred, and fast walking speeds, respectively (i.e., average total of number of 1440, 1665, and 1836 gait cycles). Note that vertical ground reaction force during walking was characterized by two maxima and one local minimum. Sagittal ankle movement, in contrast, was characterized by two dorsiflexion and two plantarflexion peaks during each gait cycle. Similarly, ultrasound transmission velocity in the Achilles tendon was characterized by two maxima and two minima during each gait cycle. *Dashed lines* represent the standard error of the mean.

dorsiflexion torque. The decrease in ankle joint dorsiflexion, however, was small (<1°) and was accompanied by an increase in ankle plantarflexion (≈1°); resulting in minimal overall change in total ankle range of motion with increasing walking speed. It is questionable, therefore, whether such small changes in ankle movement are clinically meaningful.

Second, increasing walking speed has previously been suggested to hasten the shortening velocity of gastrocnemius and soleus muscle fibers, thereby lowering the force generating ability of these muscles at faster gait speeds (4,11,25). Hence, our *in vivo* data tend to support previous research suggesting that human triceps surae muscle may suffer from less favorable contractile conditions with increasing walking speed and is unable to produce the same force as at slower walking speeds (11,25). Our findings, however, differ from these previous studies in several important ways. In Neptune and Sasaki's (25) simulations, force in the triceps surae muscle was only reduced at walking speeds faster than 80% of the walk–run transition speed (i.e., ≈1.57 $m \cdot s^{-1}$), which coincided with measurable reduction in the second "propulsive" vertical ground reaction force peak. Whereas in the experimental study of Farris and Sawicki (11), average fascicle force of the medial gastrocnemius muscle appeared to be greatest at preferred walking speed (1.25 $m \cdot s^{-1}$) and was only significantly reduced at notably slower (0.75 $m \cdot s^{-1}$) and faster (2.0 $m \cdot s^{-1}$) walking speeds. In contrast, we noted a progressive decrease in the transmission velocity of ultrasound in the Achilles tendon over a relatively narrow range of walking speeds (0.85–1.35 $m \cdot s^{-1}$), despite an increase in peak vertical ground reaction force. Although musculoskeletal simulation has suggested that the speed-induced reduction in muscle contractibility may be counteracted by the greater activation of the triceps surae muscles which occurs with increased walking speed (4), it should be noted that we did not use a measure of muscle activity in our study design. Nonetheless, our findings tend to suggest that muscle activation was not sufficient to counteract any reduction in force generation of the triceps surae muscle with increasing walking speed. Speculatively, it is possible that the reduction in ultrasound velocity in the Achilles tendon, and hence load, with increasing walking speed reflects a relatively greater contribution of passive structures (ligaments/joint capsule) or other synergist muscles, such as the long digital flexors, peroneals, or tibialis posterior to ankle plantarflexion. In support of such a concept, the coordinative strategies between ankle plantarflexor muscles (i.e., relative force contribution) have been shown to vary considerably with increasing ankle torque (12).

The observation that ultrasound velocity, and hence peak loads, in the Achilles tendon was increased at relatively slow walking speeds in this study has significant implications for the rehabilitation of Achilles tendon disorders in which progressive loading is widely advocated. Many programs typically recommend slow walking early in the management of Achilles tendon disorders to lower tendon force, with progressively faster gait speeds recommended later in the course of treatment to increase tendon loading (35). Our findings suggest that such a protocol would have the opposite effect on Achilles tendon loading. Moreover, it is noteworthy that individuals with Achilles tendon disorders, such as

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TABLE 2. Mean (SD) vertical ground reaction force parameters of participants at slow, preferred, and fast walking speeds.

	Slow	Preferred	Fast	η_p^2	P
n	33	33	33		
First ground reaction force peak (BW)	1.08 (0.05)	1.11 (0.05)	1.17* (0.07)	0.46	<0.001
Ground reaction force minimum (BW)	0.94 (0.05)	0.85 (0.08)	0.73* (0.10)	0.89	<0.001
Second ground reaction force peak (BW)	1.09 (0.04)	1.11 (0.06)	1.14* (0.09)	0.34	<0.001
Time to first ground reaction force peak (%SPD)	31.0 (3.1)	26.9 (3.0)	24.3* (3.1)	0.76	<0.001
Time to ground reaction force minimum (%SPD)	48.2 (4.1)	47.6 (3.0)	47.9 (2.4)	0.03	0.332
Time to second ground reaction force peak (%SPD)	70.6 (3.5)	73.7 (2.0)	75.6* (2.0)	0.65	<0.001
Peak loading rate ($BW \cdot s^{-1}$)	21.0 (5.1)	30.0 (7.4)	40.4* (9.6)	0.92	<0.001

*Statistically significant difference between all walking speeds ($P < 0.05$).
BW, normalized to body weight; %SPD, percentage of stance phase duration.

tendinopathy or surgically repaired tendon, naturally adopt slower walking speeds than healthy individuals (18). Based on the findings of our study, this would increase tensile loading within the Achilles tendon, which may be beneficial, given that heightened tendon loading has been shown to yield positive structural, biochemical, and biomechanical effects on tendon (1,6,20), and that heavy slow resistance training and eccentric overload protocols have been shown to be of clinical benefit in the management of Achilles tendinopathy (5). Although small, it should be noted that the magnitude of the decrease in peak ultrasound velocity in the Achilles tendon at each walking speed in our study (≈20–23 $m \cdot s^{-1}$) is comparable to the increase in peak transmission velocity reported with a change to shod walking (+25 $m \cdot s^{-1}$) and the reduction associated with inclusion of 12-mm orthotic heel lift (−14 $m \cdot s^{-1}$) (38,40). Thus, the observed change in peak ultrasound transmission velocity with increasing speed in the current study is likely to be clinically significant.

The current study has a number of limitations. Although tendons are the functional continuation of muscles, and measures of ultrasound transmission velocity in tendon provide an indication of the force developed in muscle–tendon unit as a whole, it should be noted that the Achilles tendon is a conjoined tendon of the soleus and medial and lateral gastrocnemius muscles, and the ultrasound technique used in this study is currently unable to identify differential loading within the tendon produced by these muscles (3). In contrast, transmission-mode ultrasound, as used in the current study, is neither predicated on estimates of the tendon moment arm, nor requires assumptions regarding the contribution of agonist and antagonist muscles to the net ankle joint moment, which have been shown to induce substantial errors in estimates of tendon loading (14). Rather the axial transmission speed of ultrasound is primarily related to the density and instantaneous elastic modulus of tendon. Although it can be reasonably assumed that density of tendon remains relatively stable over a 60-s data capture period (2), the instantaneous elastic modulus of tendon is dependent on the applied tensile load, which reflects a combination of both active and passive loading. Although measures of muscle activity are not without limitation (10), incorporation of synchronous measures of electromyographic activity with

TABLE 3. Mean (SD) ankle kinematics of participants at slow, preferred and fast walking speeds.

	Slow	Preferred	Fast	η_p^2	P
n	33	33	33		
Peak ankle plantarflexion 1 (°)	−5.6 (1.5)	−5.4 (1.5)	−5.2* (1.6)	0.19	0.004
Peak ankle dorsiflexion 1 (°)	5.9 (1.8)	5.6 (1.7)	5.4** (1.8)	0.25	0.001
Peak ankle plantarflexion 2 (°)	−6.7 (3.2)	−7.4 (3.3)	−7.8** (3.3)	0.39	<0.001
Peak ankle dorsiflexion 2 (°)	1.1 (1.0)	1.4 (1.0)	1.7** (1.1)	0.27	0.001
Time of peak ankle plantarflexion 1 (%GC)	9.0 (3.3)	9.1 (3.5)	9.7 (3.7)	0.02	0.462
Time of peak ankle dorsiflexion 1 (%GC)	47.8 (4.7)	45.5 (5.0)	44.4** (5.5)	0.21	0.002
Time of peak ankle plantarflexion 2 (%GC)	68.5 (5.3)	66.5 (3.1)	65.9*** (2.9)	0.15	0.010
Time of peak ankle dorsiflexion 2 (%GC)	89.3 (5.1)	89.3 (4.6)	90.2 (5.4)	0.04	0.301

*Statistically significant difference between all walking speeds ($P < 0.05$), except preferred and slow walking.
**Statistically significant difference between all walking speeds ($P < 0.05$).
***Statistically significant difference between all walking speeds ($P < 0.05$), except preferred and fast walking.

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TABLE 4. Mean (SD) ultrasound transmission velocity in the Achilles tendon at slow, preferred, and fast walking speeds.

	Slow	Preferred	Fast	η_p^2	P
n	33	33	33		
Minimum ultrasound velocity 1 ($m \cdot s^{-1}$)	2017 (154)	2001 (147)	1978* (159)	0.22	0.002
Peak ultrasound velocity 1 ($m \cdot s^{-1}$)	2255 (132)	2232 (134)	2212* (145)	0.20	0.007
Minimum ultrasound velocity 2 ($m \cdot s^{-1}$)	1943 (166)	1941 (140)	1925 (151)	0.03	0.384
Peak ultrasound velocity 2 ($m \cdot s^{-1}$)	2129 (141)	2119 (138)	2102 (146)	0.08	0.109
Time of minimum ultrasound velocity 1 (%GC)	9.5 (5.0)	8.7 (3.0)	8.8 (3.2)	0.02	0.520
Time of peak ultrasound velocity 1 (%GC)	50.2 (4.3)	49.1 (4.5)	48.6 (5.1)	0.06	0.162
Time of minimum ultrasound velocity 2 (%GC)	71.2 (6.1)	68.6 (4.6)	68.3** (5.0)	0.15	0.014
Time of peak ultrasound velocity 2 (%GC)	95.1 (3.7)	95.3 (4.2)	95.5 (4.8)	0.01	0.724

*Statistically significant difference between all walking speeds ($P < 0.05$).
**Statistically significant difference between all walking speeds ($P < 0.05$), except preferred and fast walking.

ultrasonic transmission in future research may aid efforts to discriminate between active and passive sources of tensile load in the Achilles tendon. A further caveat to this study is that we evaluated the influence of walking speed in healthy young adults during barefoot treadmill walking over a relatively narrow range of steady-state walking speeds. Although walking speeds evaluated in our study are inclusive of the range of values reported for individuals with Achilles tendon disorders and postrupture repair, the findings may not be directly transferable to individuals with Achilles tendon complaints, in which impaired triceps surae muscle activity has been reported (33). Similarly, although the instrumented treadmill system used in the current study allowed for the collection of gait data over a large number of steps and at consistent, reproducible speeds, only the vertical component of the ground reaction force could be determined. Moreover, treadmill systems are known to place spatial and temporal constraints on walking, which have been shown by some, to alter neuromuscular coordination and subsequent lower extremity joint moments and powers during walking (22). Others, however, have shown that treadmill walking has negligible effect on fascicle behavior of the gastrocnemius muscle when compared with overground walking (9). Nonetheless, the findings of this study may not necessarily be representative of unconstrained, non–steady-state or shod walking in "real-world" settings outside of the laboratory.

In conclusion, despite an increase in peak vertical ground reaction force and peak ankle plantarflexion angle, a progressive reduction in the axial transmission velocity of ultrasound in the Achilles tendon was observed with increasing walking speed (0.85–1.35 $m \cdot s^{-1}$); indicating a speed-dependent reduction in tensile load within the triceps surae muscle–tendon unit. These findings question the rationale behind current progressive loading protocols for disorders of the Achilles tendon, in which slow walking is advocated early in the course of treatment to lower peak tensile load in the tendon.

The authors would like to thank Mr Mathias Wulf for his assistance with data collection and Drs Sue Hooper and James Smeathers for their editorial contribution to the manuscript. S. C. W. is funded through an Accelerate Research Fellowship, Department of Science, Information Technology, and Innovation, Queensland Government. This research received financial support from the Australia-Germany Research Cooperation Scheme. The authors declare no conflicts of interest, financial or otherwise. The results of the present study do not constitute endorsement by the ACSM. The results of the study are presented clearly, honestly, and without fabrication, falsification, or inappropriate data manipulation.

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